

Project Proposal:

Departmental Store Data-base Management System

Project Title

Departmental Store Database Management System

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Project Description

The Departmental Store Database Management System is a database project designed to store, manage, and organize all the information related to a departmental store. The system will maintain records of products, customers, suppliers, sales, employees, and inventory in a structured database. The main purpose of this project is to replace manual record-keeping with a computerized database system that ensures data accuracy, fast access, and efficient management of store operations.

What We Are Going to Build

We are going to build a relational database system for a departmental store using SQL and a Database Management System (DBMS) such as MySQL or Oracle. The database will include multiple connected tables to manage:

- Product details
- Customer records
- Sales transactions
- Supplier information
- Employee data
- Inventory stock

Relationships between tables will help maintain data consistency and reduce redundancy.

Project Key Features

1. Product Management

- Store product information
- Product ID, name, category, price, quantity
- Update and delete product records

2. Customer Management

- Store customer details
- Customer purchase history
- Contact information management

3. Sales Management

- Record daily sales transactions
- Generate sales history
- Store invoice information

4. Inventory Management

- Track available stock
- Update stock after sales
- Low stock monitoring

5. Supplier Management

- Store supplier information
- Track supplied products
- Manage supplier contacts

6. Employee Records

- Maintain employee details
- Store employee roles and salaries

Expected Outcome

A structured, scalable database that eliminates data duplication, ensures data integrity through proper relationships, and constraints , and supports business decisions through accurate , queryable data resulting in faster checkout, better stock control and improved supplier relationships.

Code :

- **First we create Table:**

```
create table customer (  
    customer_id int primary key,  
    first_name varchar(50),  
    middle_name varchar(50),  
    last_name varchar(50),  
    phone varchar(20),  
    email varchar(100),  
    city varchar(50),  
    zip_code varchar(20)  
);
```

```
create table department (  
    dept_id int primary key,  
    dept_name varchar(100),  
    dept_location varchar(100)  
);
```

```
create table employee (  
    emp_id int primary key,
```

```
emp_name varchar(100),
salary decimal(10,2),
shift_time varchar(50),
skills varchar(100),
years_experience int,
dept_id int,
foreign key (dept_id) references department(dept_id)
);
```

```
create table product (
    product_id int primary key,
    product_name varchar(100),
    category varchar(50),
    brand varchar(50),
    price decimal(10,2),
    reorder_level int,
    dept_id int,
    foreign key (dept_id) references department(dept_id)
);
```

```
create table inventory (
    inv_id int primary key,
    product_id int,
    location varchar(100),
    last_updated date,
    quantity_available int,
```

```
foreign key (product_id) references product(product_id)  
);
```

```
create table supplier (  
    supp_id int primary key,  
    supp_name varchar(100),  
    country varchar(50),  
    rating int,  
    contact_email varchar(100)  
);
```

```
create table purchase_order (  
    po_number int primary key,  
    supp_id int,  
    order_date date,  
    total_cost decimal(10,2),  
    delivery_date date,  
    foreign key (supp_id) references supplier(supp_id)  
);
```

```
create table product_supplier (  
    product_id int,  
    supp_id int,  
    primary key (product_id, supp_id),  
    foreign key (product_id) references product(product_id),  
    foreign key (supp_id) references supplier(supp_id)
```

);

create table sales_order (

order_id int primary key,

customer_id int,

emp_id int,

order_date date,

status varchar(50),

total_amount decimal(10,2),

foreign key (customer_id) references customer(customer_id),

foreign key (emp_id) references employee(emp_id)

);

create table order_item (

item_no int primary key,

order_id int,

product_id int,

quantity int,

unit_price decimal(10,2),

line_total decimal(10,2),

foreign key (order_id) references sales_order(order_id),

foreign key (product_id) references product(product_id)

);

create table billing (

bill_id int primary key,

```
order_id int,  
customer_id int,  
emp_id int,  
payment_method varchar(50),  
invoice_code varchar(50),  
amount_paid decimal(10,2),  
bill_date date,  
foreign key (order_id) references sales_order(order_id),  
foreign key (customer_id) references customer(customer_id),  
foreign key (emp_id) references employee(emp_id)  
);
```

Now we insert data in tables

insert into customer values

```
(1, 'Ali', 'Raza', 'Khan', '03001234567', 'ali@gmail.com', 'Lahore', '54000'),  
(2, 'Sara', 'Noor', 'Ahmed', '03111234567', 'sara@gmail.com', 'Karachi', '75000'),  
(3, 'Hassan', 'Ali', 'Malik', '03221234567', 'hassan@gmail.com', 'Islamabad',  
'44000'),  
(4, 'Ayesha', 'Bibi', 'Rauf', '03331234567', 'ayesha@gmail.com', 'Lahore',  
'54000');
```

insert into department values

```
(1, 'Grocery', 'Ground Floor'),  
(2, 'Electronics', 'First Floor'),
```


(3, 'Clothing', 'Second Floor');

insert into employee values

(1, 'Ahmed Shah', 45000, 'Morning', 'Billing', 3, 1),

(2, 'Usman Ali', 55000, 'Evening', 'Sales', 5, 2),

(3, 'Fatima Noor', 50000, 'Morning', 'Inventory', 4, 3);

insert into product values

(1, 'Rice Bag', 'Grocery', 'Falak', 2500, 10, 1),

(2, 'LED TV', 'Electronics', 'Samsung', 85000, 3, 2),

(3, 'Men Shirt', 'Clothing', 'Outfitters', 3000, 15, 3),

(4, 'Cooking Oil', 'Grocery', 'Dalda', 1200, 20, 1);

insert into inventory values

(1, 1, 'Rack A1', '2026-06-10', 50),

(2, 2, 'Rack B2', '2026-06-10', 8),

(3, 3, 'Rack C3', '2026-06-10', 40),

(4, 4, 'Rack A2', '2026-06-10', 60);

insert into supplier values

(1, 'Metro Suppliers', 'Pakistan', 5, 'metro@gmail.com'),

(2, 'Tech World', 'China', 4, 'techworld@gmail.com'),

(3, 'Fashion Hub', 'Pakistan', 4, 'fashionhub@gmail.com');

insert into purchase_order values

(101, 1, '2026-06-01', 50000, '2026-06-05'),
(102, 2, '2026-06-02', 170000, '2026-06-06'),
(103, 3, '2026-06-03', 60000, '2026-06-07');

insert into product_supplier values

(1, 1),
(4, 1),
(2, 2),
(3, 3);

insert into sales_order values

(1, 1, 1, '2026-06-10', 'Completed', 6200),
(2, 2, 2, '2026-06-10', 'Completed', 85000),
(3, 3, 3, '2026-06-10', 'Pending', 6000);

insert into order_item values

(1, 1, 1, 2, 2500, 5000),
(2, 1, 4, 1, 1200, 1200),
(3, 2, 2, 1, 85000, 85000),
(4, 3, 3, 2, 3000, 6000);

insert into billing values

(1, 1, 1, 1, 'Cash', 'INV-001', 6200, '2026-06-10'),
(2, 2, 2, 2, 'Card', 'INV-002', 85000, '2026-06-10'),
(3, 3, 3, 3, 'JazzCash', 'INV-003', 3000, '2026-06-10');

Now we are going to data which we stored in tables

```
select * from customer;  
select * from department;  
select * from employee;  
select * from product;  
select * from inventory;  
select * from supplier;  
select * from purchase_order;  
select * from product_supplier;  
select * from sales_order;  
select * from order_item;  
select * from billing;
```

10 Project Features:

1. Add New Customer

insert into customer values

(5, 'Bilal', 'Ahmad', 'Sheikh', '03441234567', 'bilal@gmail.com', 'Multan', '60000');

2. Add new Product

insert into product values

```
(5, 'Mobile Phone', 'Electronics', 'Oppo', 65000, 5, 2);
```

3. Search Customer By City

```
select *  
from customer  
where city = 'Lahore';
```

4. Display products with department name

```
select  
    product.product_id,  
    product.product_name,  
    product.category,  
    product.brand,  
    product.price,  
    department.dept_name  
from product  
join department ON product.dept_id = department.dept_id;
```

5. Display sales orders with customer and employee details

```
select  
    sales_order.order_id,  
    customer.first_name,
```

```
customer.last_name,  
employee.emp_name,  
sales_order.order_date,  
sales_order.status,  
sales_order.total_amount  
from sales_order  
join customer  
ON sales_order.customer_id = customer.customer_id  
join employee  
ON sales_order.emp_id = employee.emp_id;
```

6. Display order items with product details

```
select  
    order_item.item_no,  
    sales_order.order_id,  
    product.product_name,  
    order_item.quantity,  
    order_item.unit_price,  
    order_item.line_total  
from order_item  
join sales_order  
on order_item.order_id = sales_order.order_id  
join product  
on order_item.product_id = product.product_id;
```

7. Update product price

```
update product
set price = 2700
where product_id = 1;
```

8. Update order status

```
update sales_order
set status = 'Completed'
where order_id = 3;
```

9. Delete one customer record

```
delete from customer
where customer_id = 5;
```

10. Display low stock products

```
select
    product.product_id,
    product.product_name,
    inventory.quantity_available,
    product.reorder_level
from product
join inventory
on product.product_id = inventory.product_id
```

where inventory.quantity_available <= product.reorder_level;

11. Display supplier with supplied products

```
select
    supplier.supp_name,
    product.product_name,
    product.brand,
    product.price
from supplier
join product_supplier
on supplier.supp_id = product_supplier.supp_id
join product
on product_supplier.product_id = product.product_id;
```

12. Display billing with customer and order details

```
select
    billing.bill_id,
    billing.invoice_code,
    customer.first_name,
    customer.last_name,
    billing.payment_method,
    billing.amount_paid,
```

```
    billing.bill_date
from billing
join customer
on billing.customer_id = customer.customer_id;
```

13. Display total sales amount

```
select sum(total_amount) as total_sales
from sales_order;
```

14. Display highest price product

```
select *
from product
where price = (
    select max(price)
    from product
);
```

15. Display employee department details

```
select
    employee.emp_id,
    employee.emp_name,
    employee.salary,
    department.dept_name,
```



```
department.dept_location  
from employee  
join department  
on employee.dept_id = department.dept_id;
```

Git Hub Link

Ameer Hamza Repo Link

<https://github.com/Ameeri110/DB-LAB-PROJECT-.git>

Muhammad Abdulwahab Repo Link

<https://github.com/AbdulWahabsultan/Data-base-uol>

Asadullah Aslam Repo Link

<https://github.com/Asadullah567/DB-Lab-Project.git>

Zeeshan Ali

<https://github.com/70154127-bit/Zeeshan-Ali>